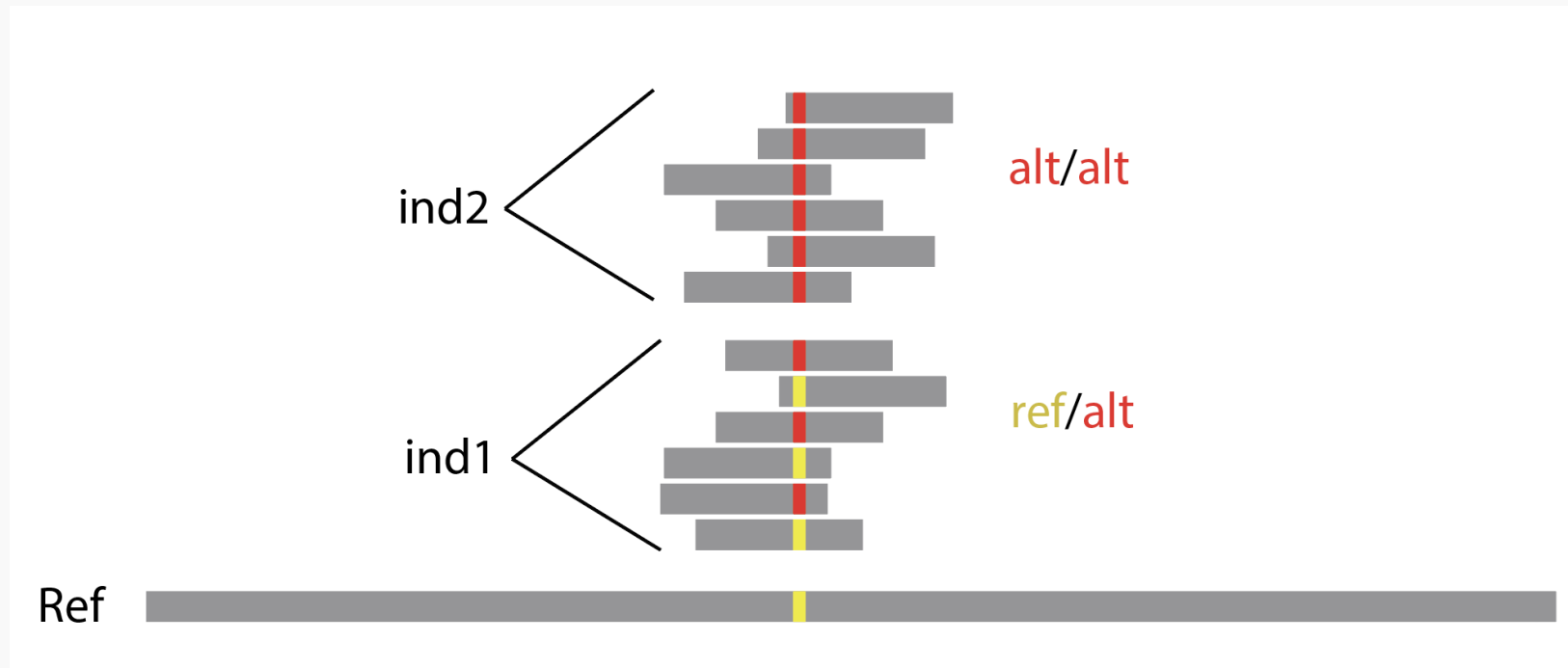


3. Multi-sample variant calling with BCFtools

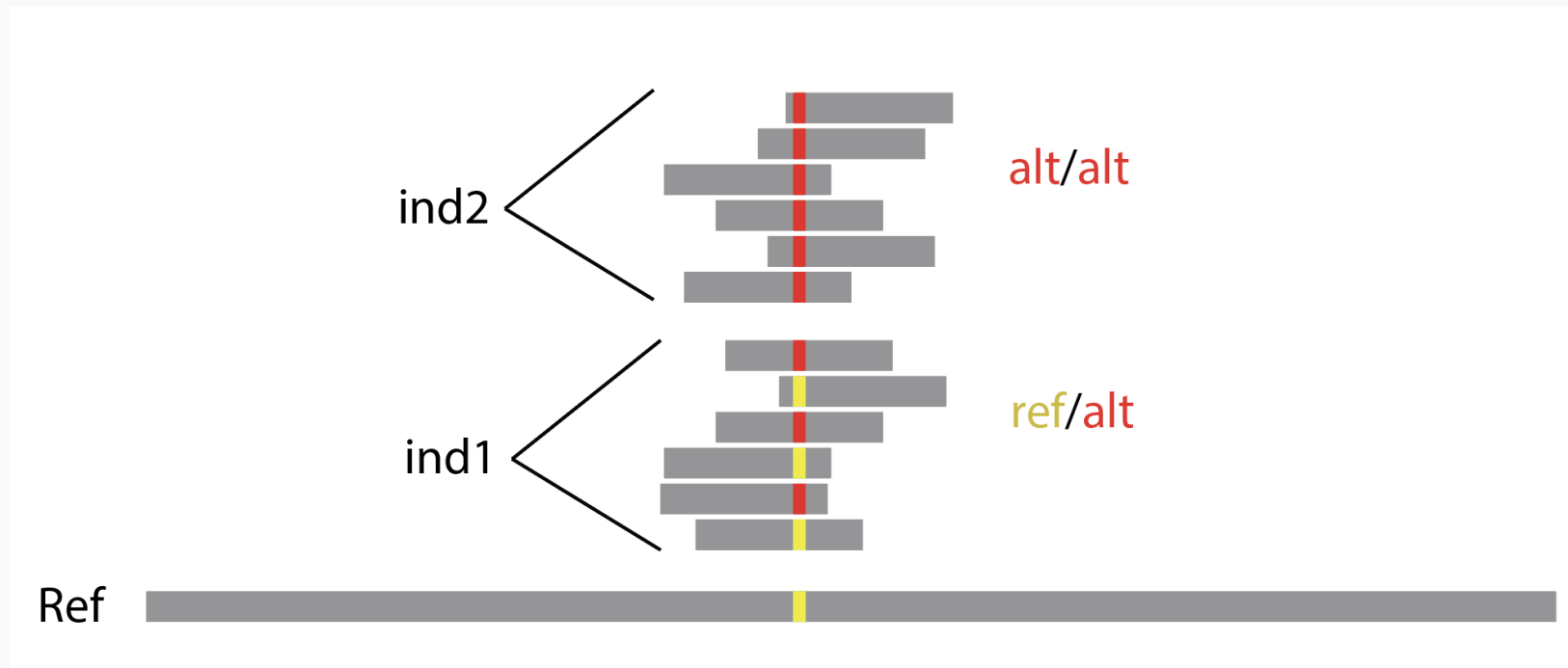
Variant calling

The process of inferring genetic differences from sequencing reads relative to a reference genome:



Variant calling

The process of inferring genetic differences from sequencing reads relative to a reference genome:



Variant calling

Each individual carries 2 alleles, and we want to infer the state of both from n reads.



How random flips would you want to confidently conclude that this is not a 2-headed coin?

The reality is that you can never be 100% certain

Variant calling

Each individual carries 2 alleles, and we want to infer the state of both from n reads.

$$P_{\text{both alleles}} = 1 - 2^{1-n}$$

where n is the number of reads

```
reads <- 1:20
prob_both <- 1 - 2^(1 - reads)
data.frame(
  reads = reads,
  probability = prob_both,
  percentage = 100 * prob_both
)
```

Reads	Probability of observing both alleles
1	0.00%
2	50.00%
3	75.00%
4	87.50%
5	93.75%
6	96.88%
7	98.44%
8	99.22%
9	99.61%
10	99.80%
11	99.90%
12	99.95%
13	99.98%
14	99.99%
15	99.994%

Variant calling

How many heterozygous genotypes would we would expect to miss-call with 5x coverage?

@ 5x coverage, we have a 93.75% chance of observing both alleles

Genome-wide heterozygosity in *Littorina* is ~0.5% and genome size is 1.35 gbp, so we expect 6.5 million heterozygous calls per individual

$$(1-93.75) * 6,500,000 = 406,250...$$

Variant calling can be informed by the ‘population’

Hardy–Weinberg equilibrium can give a prior expectation for genotype frequencies when variants are called jointly across individuals.

- If the estimated alternative allele frequency is p , then

$$P(AA) = (1 - p)^2, \quad P(Aa) = 2p(1 - p), \quad P(aa) = p^2$$

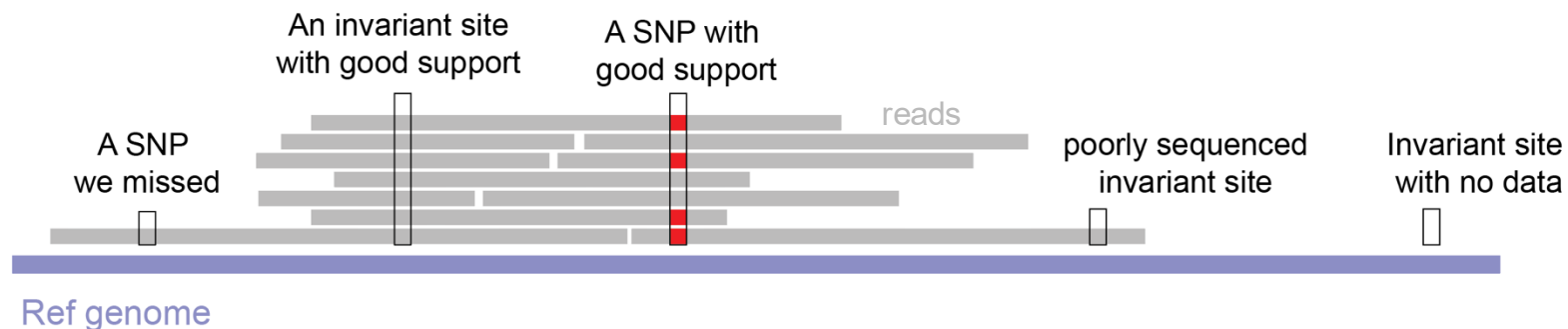
- The caller combines the evidence from the sequencing reads with the expected frequency of each genotype to estimate which genotype is most likely.
- The effect of the prior is strongly dependent on the number of reads that support a genotype.

Its not all about variants!

- The default for most callers is to only write variant sites to the output
- Many analyses require information from both variant and invariant sites
- We cannot assume that sites not called as variants are invariant
- e.g., a site not in our variant-only VCF may have been poorly sequenced, rather than showing strong evidence for being monomorphic

Its not all about variants!

- The default for most callers is to only write variant sites to the output
- Many analyses require information from both variant and invariant sites
- We cannot assume that sites not called as variants are invariant
- e.g., a site not in our variant-only VCF may have been poorly sequenced, rather than showing strong evidence for being monomorphic



Its not all about variants!

- The default for most callers is to only write variant sites to the output
- Many analyses require information from both variant and invariant sites
- We cannot assume that sites not called as variants are invariant
- e.g., a site not in our variant-only VCF may have been poorly sequenced, rather than showing strong evidence for being monomorphic
- The same level of evidence must be used for identifying variant or invariant sites
- For many caller there is an option to emit all sites (i.e., not just variants)—use this or live to regret it!

Many callers are available!

And they have very different philosophies!

- **Pileup callers:** Evaluate each site independently using the reads aligned to it; fast but less effective for complex variation.
- **Haplotype callers:** Reconstruct local sequences to identify SNPs, indels and complex variants.
- **Machine-learning callers:** Learn to distinguish variants from errors using validated training data.
- **Genotype-likelihood approaches:** Retain uncertainty in each genotype rather than making hard calls.

Many callers are available!

And they have very different philosophies!

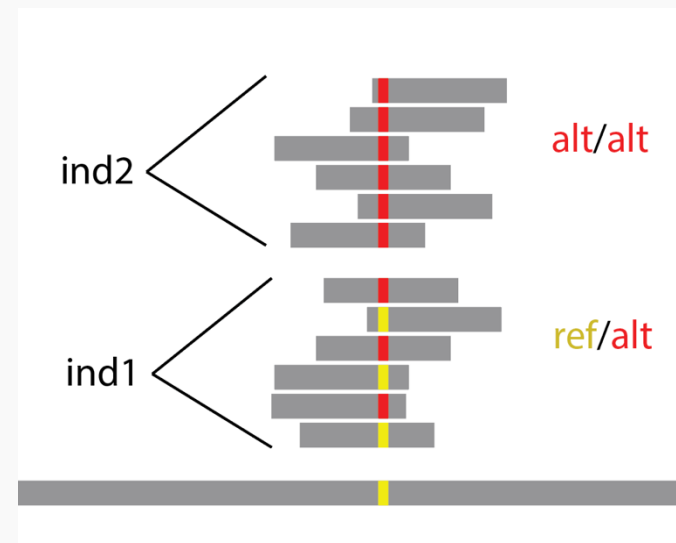
- **Pileup callers:** Evaluate each site independently using the reads aligned to it; fast but less effective for complex variation.
- **Haplotype callers:** Reconstruct local sequences to identify SNPs, indels and complex variants.

BCFtools
mpileup | call



BCFtools mpileup | call

- **Each base called independently of all others**
- **Builds a pileup:** Collects reads aligned to each position in the reference genome.
- **Weights read evidence:** Uses base and mapping qualities rather than simply counting alleles.
- **Calculates genotype likelihoods:** Estimates how well the reads support each possible genotype.
- **Calls variants across samples:** By default, treats samples as one population and applies an HWE assumption (Can be deactivated using `-G` option)





- **Reconstructs haplotypes:** Locally reassembles reads into candidate sequences rather than examining sites independently.
- **Calculates genotype likelihoods:** Uses a Pair-HMM to determine how well each haplotype explains the reads.
- **Supports joint genotyping:** Calls samples individually as gVCFs before combining evidence across samples.
- **Much easier to combined new samples with old ones:** with BCFtools, individuals need to be recalled

```
A T G T T G G T A G C T A G T T T A G C
A A G C T A G T A G C T A A T T T T G C
A A G C T A G T A G C T A A T T T T G C
A T G T T G G T A G C T A G T T T A G C
A A G C T N G T A G C T A A T T T T G C
A A G C T A G T A G C T A A T T T T G C
A T G T T G G T A G C T A G T T T A G C
A T G T T G G T A G C T A G T T T A G C
A T G T T G G T A G C T A G T T T A G C
A A G C T A G T A G C T A A T T T N G C
A A G C T A G T A G C T A A T T T T G C
A A G C T A G T A G C T A A T T T T G C
A A G C T N G T A G C T A A T T T T G C
A T G T T G G T A G C T A G T T T A G C
```



- **Reconstructs haplotypes:** Locally reassembles reads into candidate sequences rather than examining sites independently.
- **Calculates genotype likelihoods:** Uses a Pair-HMM to determine how well each haplotype explains the reads.
- **Supports joint genotyping:** Calls samples individually as gVCFs before combining evidence across samples.
- **Much easier to combined new samples with old ones:** with BCFtools, individuals need to be recalled

```
ATGTTGGTAGCTAGTTTAGC
AAGCTAGTAGCTAATTTTGC
AAGCTAGTAGCTAATTTTGC
ATGTTGGTAGCTAGTTTAGC
AAGCTAGTAGCTAATTTTGC
AAGCTAGTAGCTAATTTTGC
ATGTTGGTAGCTAGTTTAGC
ATGTTGGTAGCTAGTTTAGC
ATGTTGGTAGCTAGTTTAGC
AAGCTAGTAGCTAATTTTGC
AAGCTAGTAGCTAATTTTGC
AAGCTAGTAGCTAATTTTGC
AAGCTAGTAGCTAATTTTGC
AAGCTAGTAGCTAATTTTGC
ATGTTGGTAGCTAGTTTAGC
```

OPEN

The evaluation of Bcftools mpileup and GATK HaplotypeCaller for variant calling in non-human species

Messaoud Lefouili & Kiwoong Nam 

Identification of genetic variations is a central part of population and quantitative genomics studies based on high-throughput sequencing data. Even though popular variant callers such as Bcftools mpileup and GATK HaplotypeCaller were developed nearly 10 years ago, their performance is still largely unknown for non-human species. Here, we showed by benchmark analyses with a simulated insect population that Bcftools mpileup performs better than GATK HaplotypeCaller in terms of recovery rate and accuracy regardless of mapping software. The vast majority of false positives were observed from repeats, especially for GATK HaplotypeCaller. Variant scores calculated by GATK did not clearly distinguish true positives from false positives in the vast majority of cases, implying that hard-filtering with GATK could be challenging. These results suggest that Bcftools mpileup may be the first choice for non-human studies and that variants within repeats might have to be excluded for downstream analyses.